# Data and frequency tables

**1**

| Score | Frequency |
|---|---|
| 1 | 4 |
| 2 | 2 |
| 3 | 5 |
| 4 | 3 |
| 5 | 8 |
| 6 | 3 |

**2**

| Score$(x)$ | Frequency$(f)$ | $xf$ |
|---|---|---|
| 1 | 7 | 7 |
| 2 | 4 | 8 |
| 3 | 7 | 21 |
| 4 | 8 | 32 |
| 5 | 8 | 40 |
| 6 | 8 | 48 |

**3** 8

**4** 4.5

**5**

**a**

| Score $(x)$ | Frequency $(f)$ | $xf$ |
|---|---|---|
| 5 | 14 | 70 |
| 7 | 4 | 28 |
| 9 | 2 | 18 |
| 11 | 18 | 198 |
| 13 | 6 | 78 |
| Total | 44 | 392 |

**b** 8.91

**6**

**a** 5

**b** 72

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**7**

**a**

| Score$(x)$ | Frequency$(f)$ | $xf$ |
|---|---|---|
| 23 | 6 | 138 |
| 24 | 9 | 216 |
| 25 | 4 | 100 |
| 26 | 1 | 26 |
| 27 | 5 | 135 |
| Total | 25 | 615 |

**b** 24.6

**c** 4

**d** 24

## Frequency histograms

**8**

**a**

| Score | Frequency |
|---|---|
| 2 | 5 |
| 3 | 2 |
| 4 | 6 |
| 5 | 9 |
| 6 | 7 |
| 7 | 9 |

**b**

| Score $(x)$ | Frequency $(f)$ | $f \times x$ |
|---|---|---|
| 3 | 4 | 12 |
| 4 | 1 | 4 |
| 5 | 9 | 45 |
| 6 | 1 | 6 |
| 7 | 4 | 28 |
| 8 | 6 | 48 |

**9**

**a**

**i** 2 **ii** 4 **iii** 3.6

**b**

**i** 5 and 7 **ii** 6 **iii** 6.2

**c**

**i** 6 **ii** 6 **iii** 5.0

**d**

**i** 39 **ii** 40 **iii** 40.7

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**10**

**a**

| Score ($x$) | Frequency ($x$) | $f \times x$ |
|---|---|---|
| 2 | 6 | 12 |
| 3 | 9 | 27 |
| 4 | 4 | 16 |
| 5 | 1 | 5 |
| 6 | 5 | 30 |
| Total | 25 | 90 |

**b** 3.6

**c** 4

**d** 3

**11**

**a**

**i** 3.93 **ii** 5

**b**

**i** 43.97 **ii** $42, 44, 46$

## Frequency polygons

**12**

**a**

| Score | Frequency |
|---|---|
| 3 | 2 |
| 4 | 2 |
| 5 | 7 |
| 6 | 4 |
| 7 | 1 |
| 8 | 3 |

**b**

| Score ($x$) | Frequency ($f$) | $f \times x$ |
|---|---|---|
| 2 | 7 | 14 |
| 3 | 5 | 15 |
| 4 | 6 | 24 |
| 5 | 7 | 35 |
| 6 | 3 | 18 |
| 7 | 9 | 63 |

**13** 7

**14** 5

**15** 4.7

**16**

**a** 40.32

**b** 41

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**17** **a** 2 **b** 2 **c** 2

## Applications

**18** 2.8

**19** 14.8

**20** **a** 11 **b** 13 **c** 63 **d** 9

**21** **a** 10 **b** 18

**22** **a**

| Number of pets | Frequency |
|---|---|
| 0 | 14 |
| 1 | 17 |
| 2 | 5 |
| 3 | 9 |

**b** 45

**c** 54 **d** 1.2

**23** **a**

| Number of siblings | Frequency |
|---|---|
| 0 | 12 |
| 1 | 15 |
| 2 | 3 |
| 3 | 5 |

**b** 1

**c** 3

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